

E9: 309 ADL 4-1-2020



Housekeeping

✳ Midterm project III

- ➡ Abstract submission deadline (Jan 10th)

- ✓ Evaluation after final exam (1st week of Feb)

✳ Final Exam (as per IISc schedule)

- ✓ Jan 23rd afternoon!

Problem with current deep learning networks

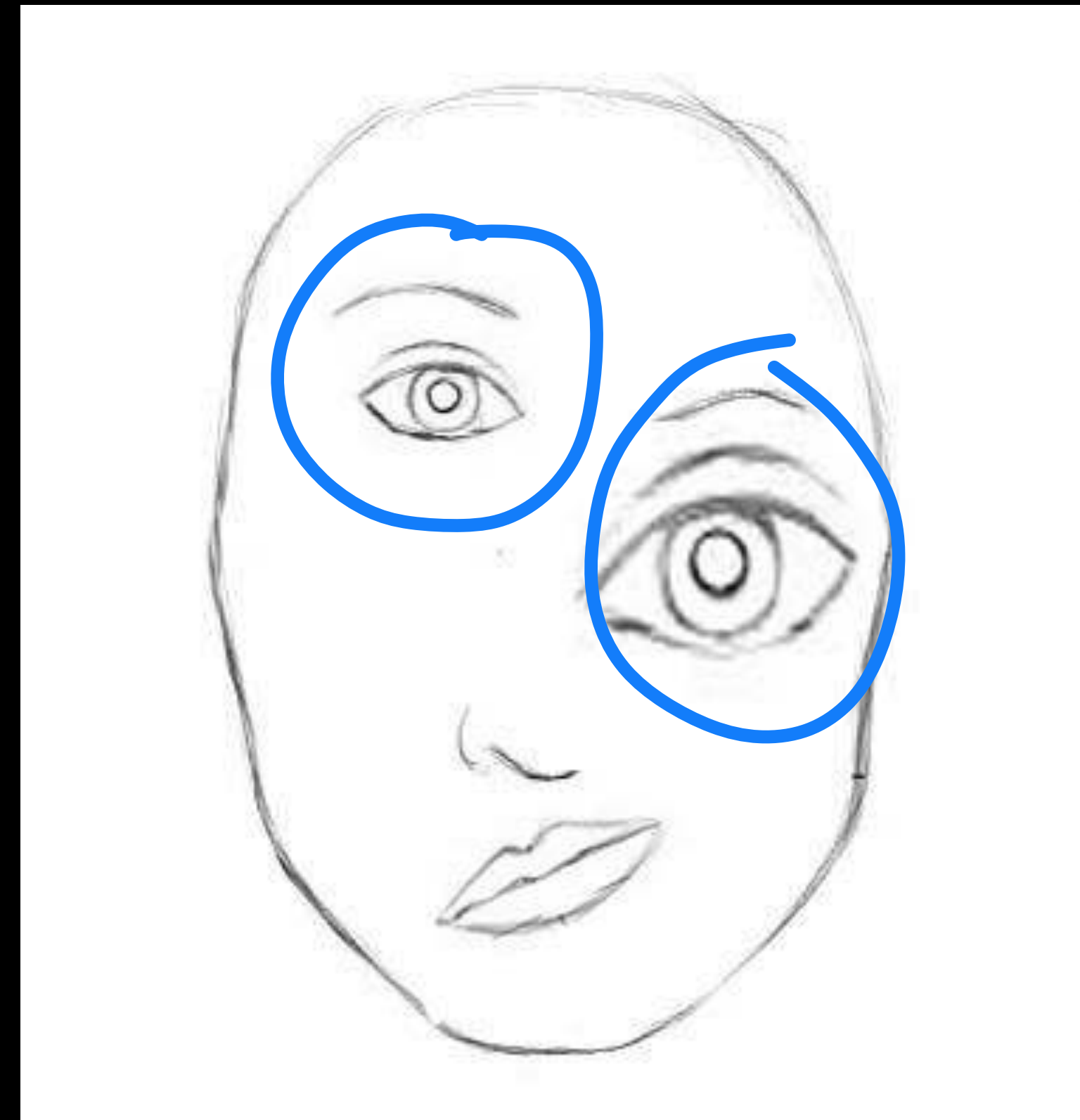
* Question

→ Can we learn the presence of object parts and their implicit spatial information (position)

* Invariance versus equivariance

→ Invariance - resistance to translations and shifts

→ Equivariance - relationship of parts with other parts have to preserved.



Is this a face ?

Capsule networks

✳ Traditional networks

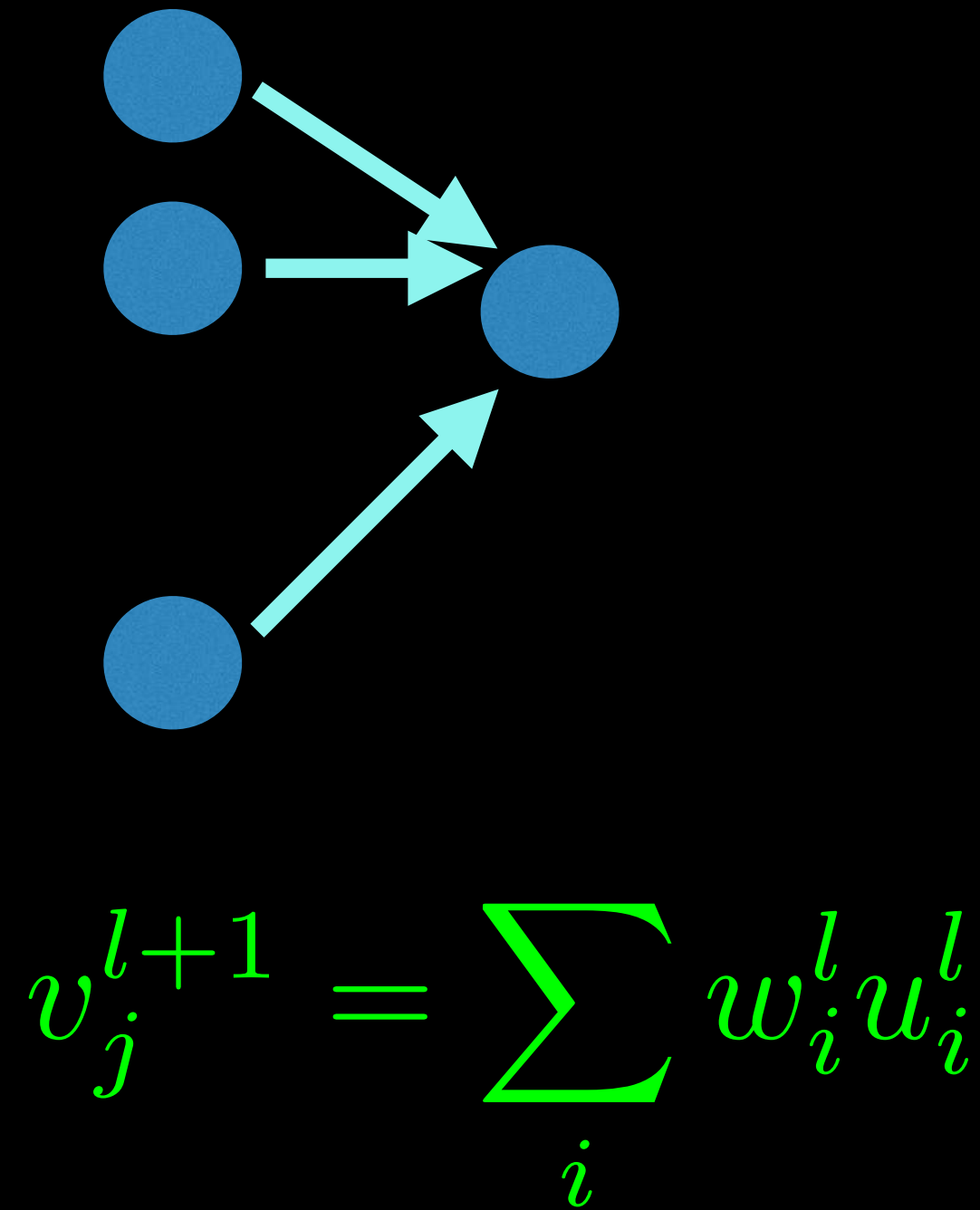
➡ Weigh the inputs and generate a scalar output

✳ Key idea in capsule networks (neurons to capsules)

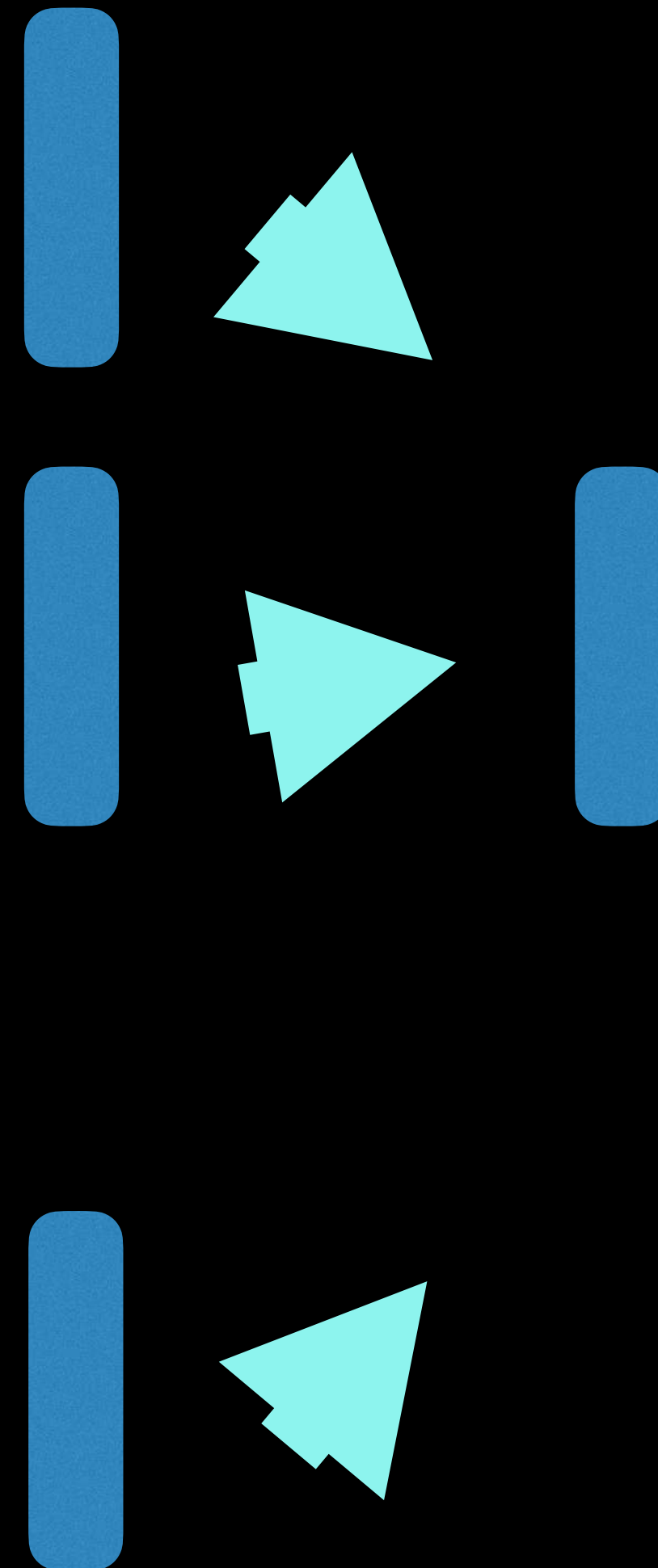
- ✓ Move individual neuron outputs from scalar to vector ✓
- ✓ Encode the probability of presence of an attribute along the magnitude of the vector output
- ✓ Encode the pose (translation + rotation) in the angle of the vector output

From a layer of neurons to layer of capsules

* Conventional neurons



* Capsule network



Prediction vector

$$\hat{\mathbf{u}}_{j|i}^{l+1} = \mathbf{W}_{ij} \mathbf{u}_i^l$$

Coupling

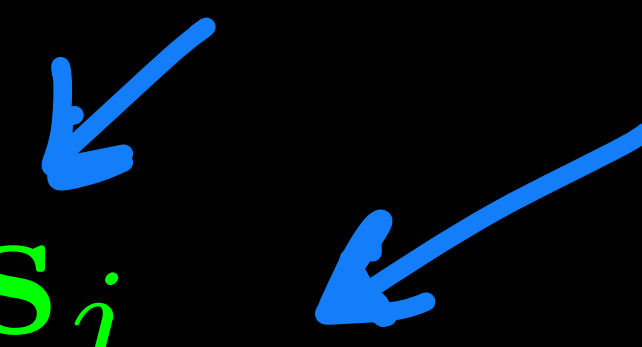
$$\mathbf{s}_j^{l+1} = \sum_i c_{ij} \hat{\mathbf{u}}_{j|i}^{l+1}$$

Instantiation parameter

$$\mathbf{v}_j^{l+1} = \text{Squash}(\mathbf{s}_j^{l+1})$$

Capsule network

* Squashing non-linearity

$$\text{Squash}(\mathbf{s}_j^{l+1}) = \frac{||\mathbf{s}_j||^2}{1 + ||\mathbf{s}_j||^2} \frac{\mathbf{s}_j}{||\mathbf{s}_j||}$$


→ Makes the vector magnitude range from (0,1)

✓ Interpretation of the length of the vector as a probability of the presence of a particular object part

→ Preserves the direction of the original vector.



Capsule network

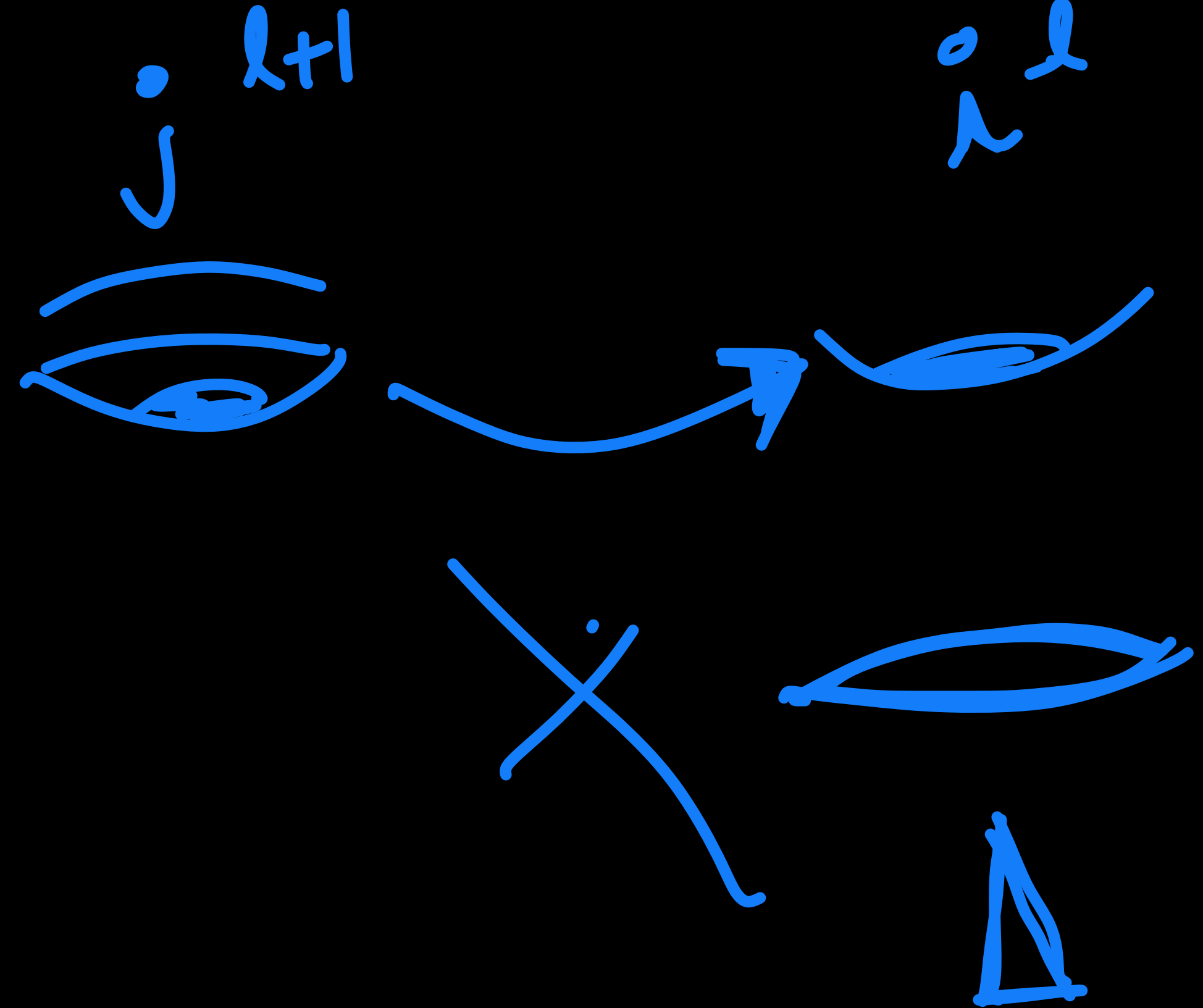
* Coupling

$$c_{ij} = \frac{e^{b_{ij}}}{\sum_k e^{b_{ik}}}$$

* The coefficients

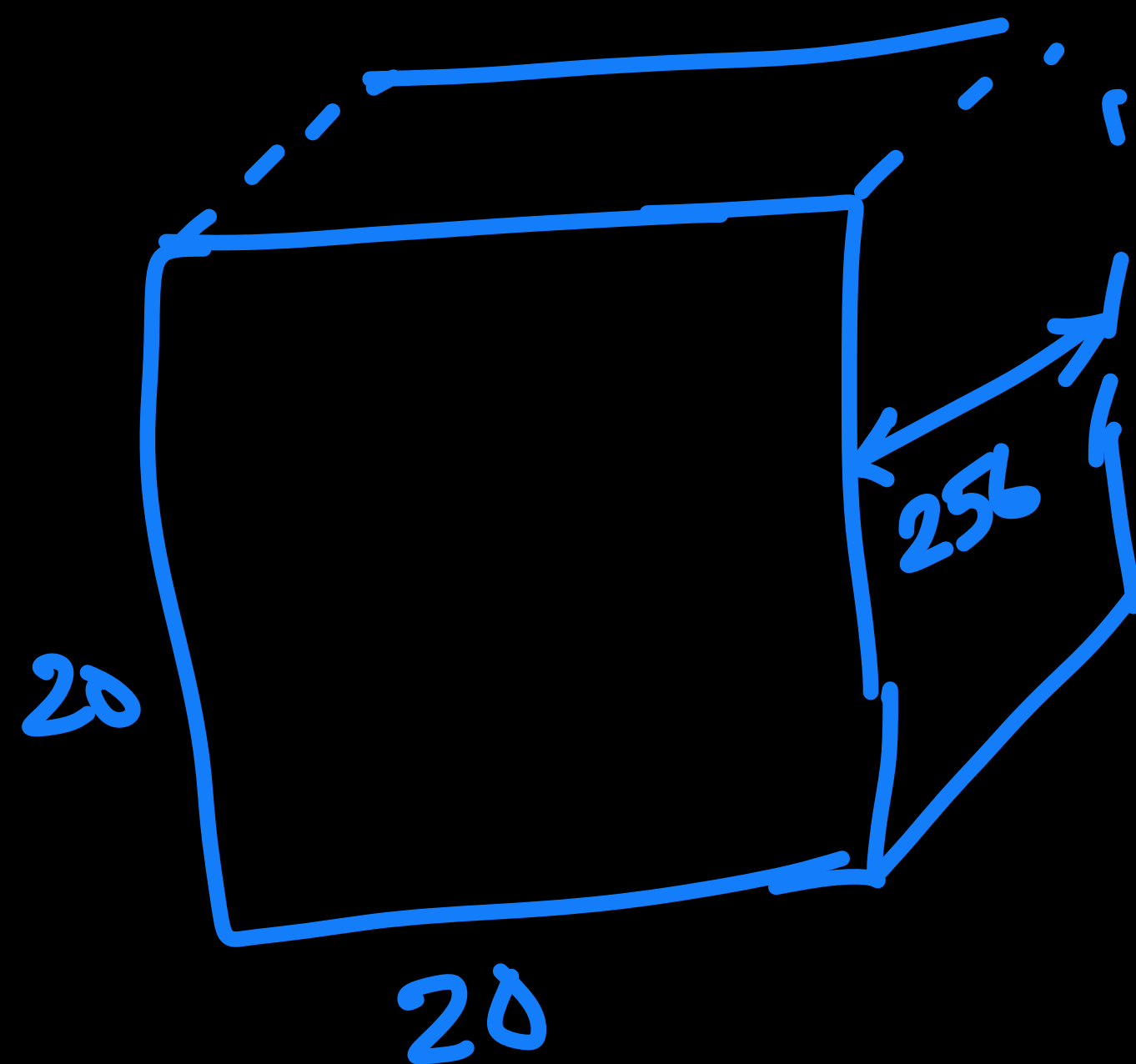
$$b_{ij} \leftarrow b_{ij} + \mathbf{u}_{j|i}^T \mathbf{v}_j$$

- ✓ The dot product measures the agreement of the layer output at j with the prediction made only based on i
- ✓ Depend on the location and type of the capsule



Neurons versus Capsules

Capsule vs. Traditional Neuron			
Input from low-level capsule/neuron		vector($\mathbf{u}_i$)	scalar(x_i)
Operation	Affine Transform	$\hat{\mathbf{u}}_{j i} = \mathbf{W}_{ij} \mathbf{u}_i$	—
	Weighting	$\mathbf{s}_j = \sum_i c_{ij} \hat{\mathbf{u}}_{j i}$	$a_j = \sum_i w_i x_i + b$
	Sum		
	Nonlinear Activation	$\mathbf{v}_j = \frac{\ \mathbf{s}_j\ ^2}{1 + \ \mathbf{s}_j\ ^2} \frac{\mathbf{s}_j}{\ \mathbf{s}_j\ }$	$h_j = f(a_j)$
Output		vector($\mathbf{v}_j$)	scalar(h_j)



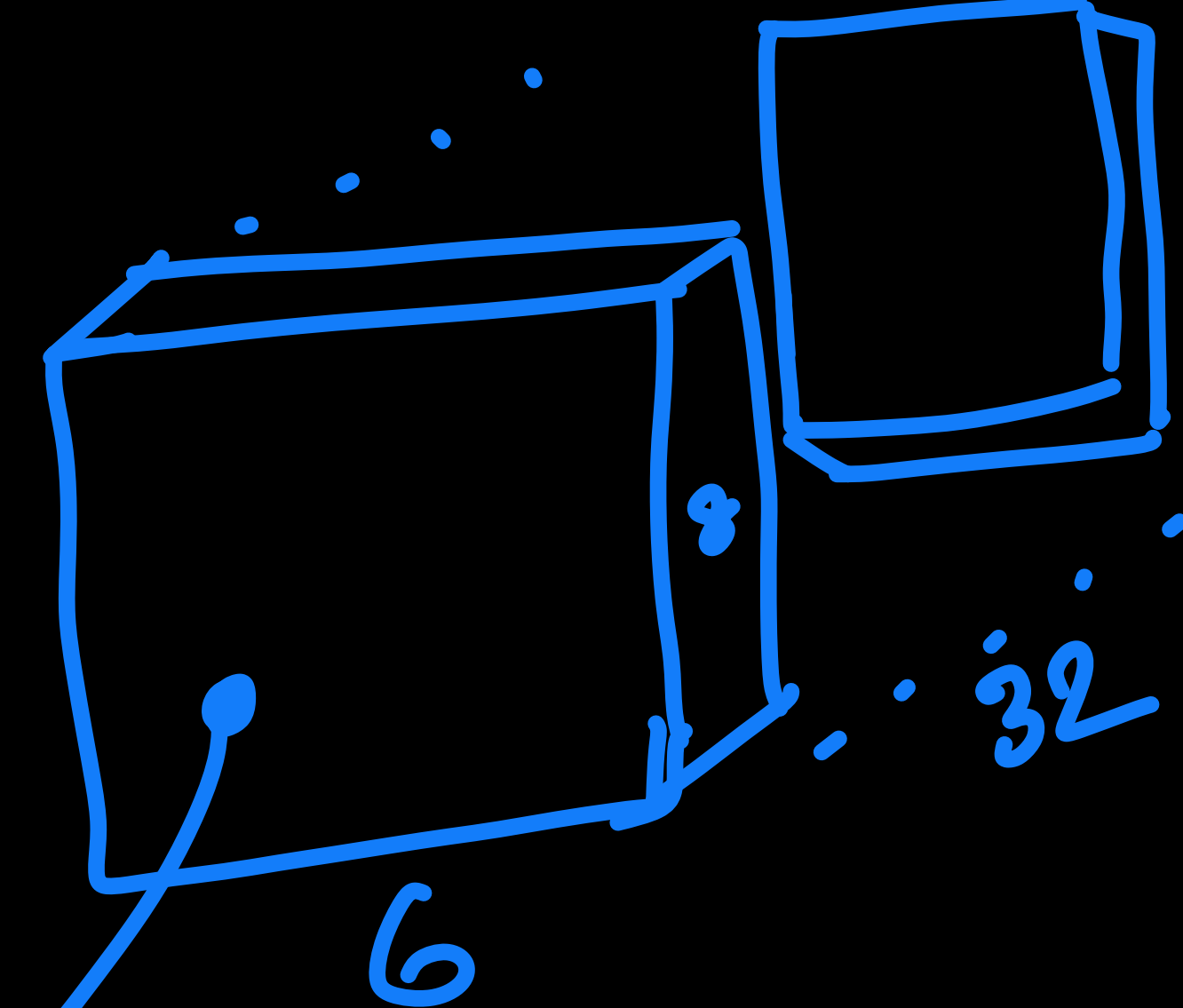
(*)

8 parallel

9×9
stride
2



6



8D.

$(32 \times 6 \times 6)$

8D

1152

0 [16D]

1:

9 [16D]



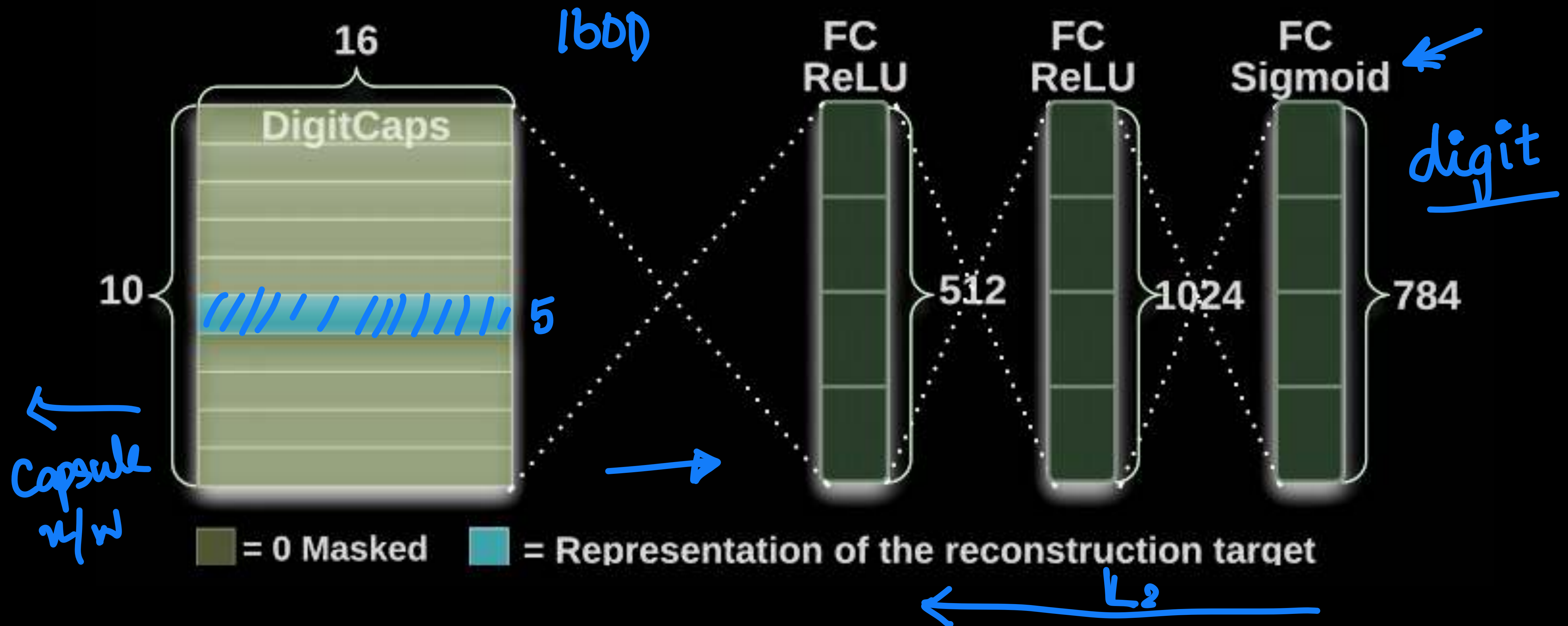
$$E = \text{Margin loss} + \lambda \text{ recon. loss}$$

←
get weights.

w_{ij}
 b_{ij}

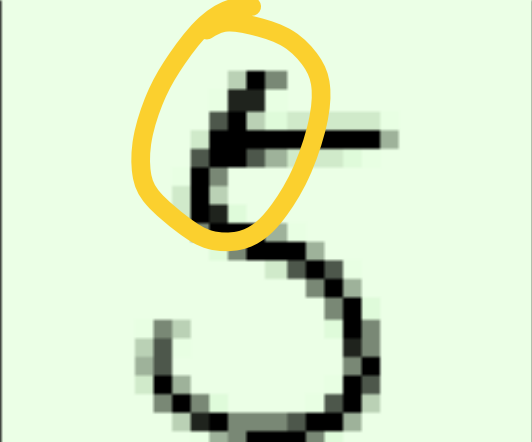
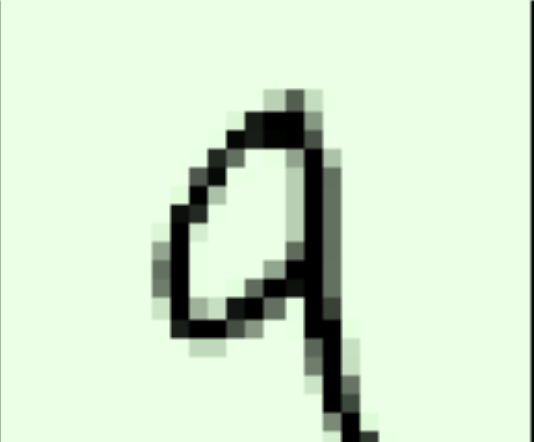
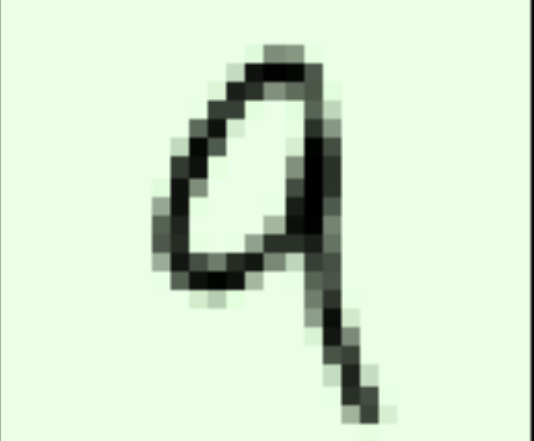
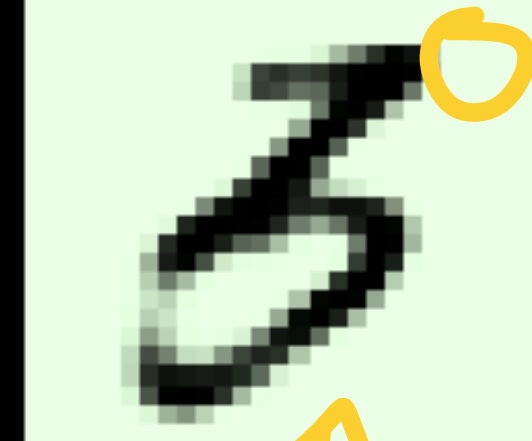
Reconstruction as a regularization loss

✳ Use the final capsules to reconstruct the digit images



Capsule networks

Figure 3: Sample MNIST test reconstructions of a CapsNet with 3 routing iterations. (l, p, r) represents the label, the prediction and the reconstruction target respectively. The two rightmost columns show two reconstructions of a failure example and it explains how the model confuses a 5 and a 3 in this image. The other columns are from correct classifications and shows that model preserves many of the details while smoothing the noise.

(l, p, r)	<u>(2, 2, 2)</u>	(5, 5, 5)	(8, 8, 8)	(9, 9, 9)	<u>5, 3, 5</u>	(5, 3, 3)
Input						
Output						

Understanding the capsule output

Figure 4: Dimension perturbations. Each row shows the reconstruction when one of the 16 dimensions in the DigitCaps representation is tweaked by intervals of 0.05 in the range $[-0.25, 0.25]$.

<u>Scale and thickness</u>	
<u>Localized part</u>	
<u>Stroke thickness</u>	
<u>Localized skew</u>	
<u>Width and translation</u>	
<u>Localized part</u>	

Capsule network performance



Table 1: CapsNet classification test accuracy. The MNIST average and standard deviation results are reported from 3 trials.

Method	Routing	Reconstruction	MNIST (%)	MultiMNIST (%)
Baseline	-	-	0.39	8.1
CapsNet	1	no	0.34 ± 0.032	-
CapsNet	1	yes	0.29 ± 0.011	7.5
CapsNet	3	no	0.35 ± 0.036	-
CapsNet	3	yes	0.25 ± 0.005	5.2

888
8

|| 84 ||
|| 4 ||

84

32 CNN + 2L FF ← 32M

CNN + 1 capsule
6M

Recons
FF n/w
2M



Disadvantage

CIFAR-10

Capsule Model tends to encode all details
v/s

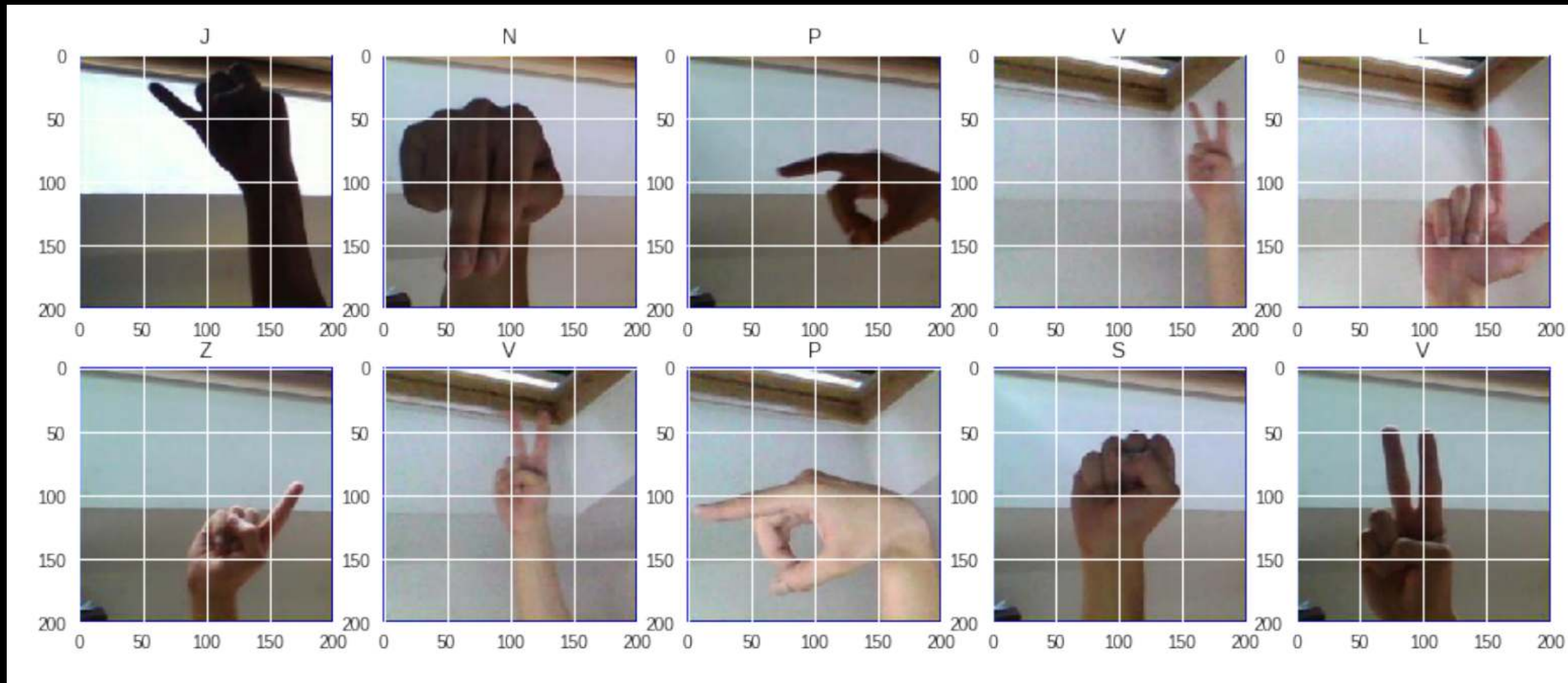
Conventional models normalize all details.

o o o o o

← reconstruction

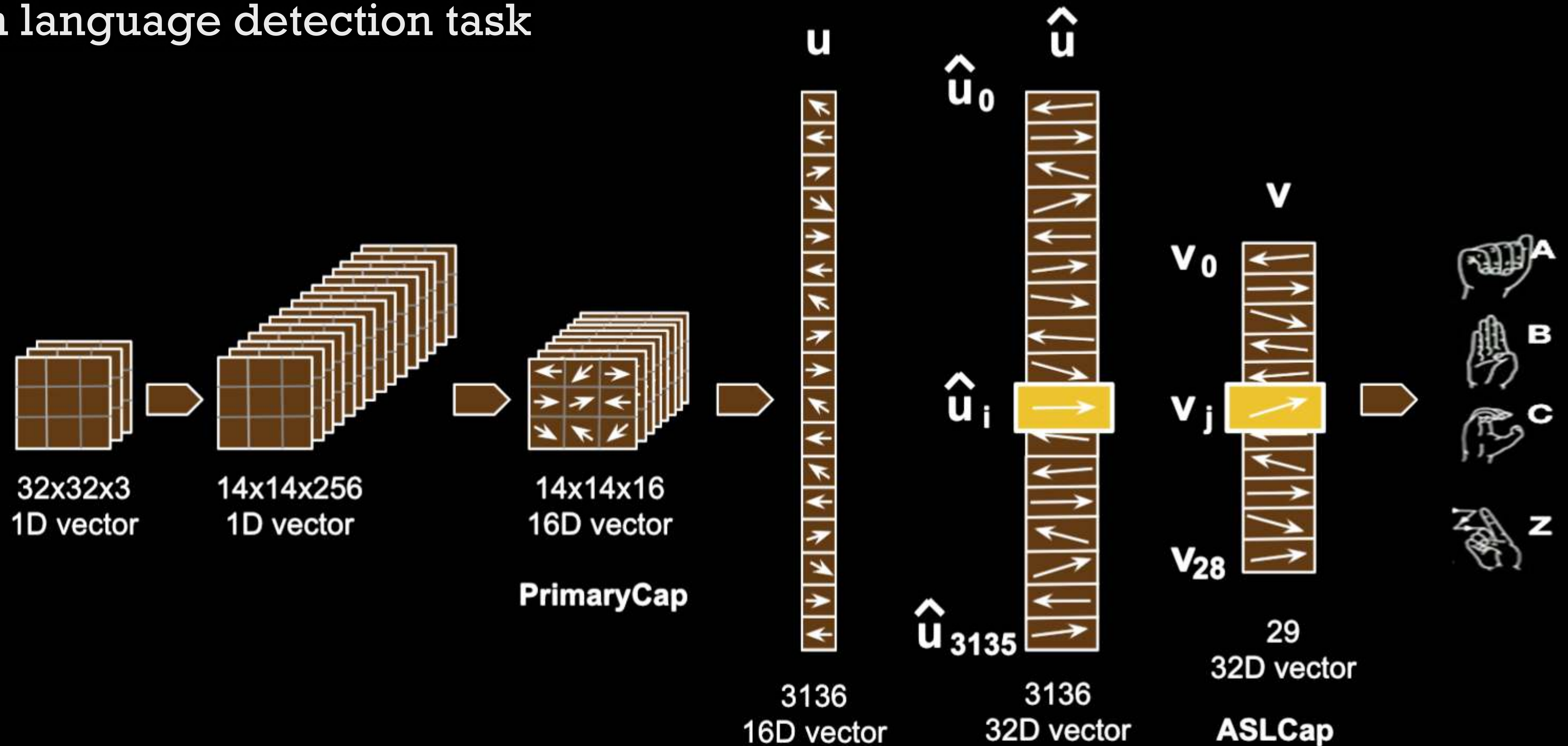
Automatic Sign Language Detection Task

✳ Sign language detection task

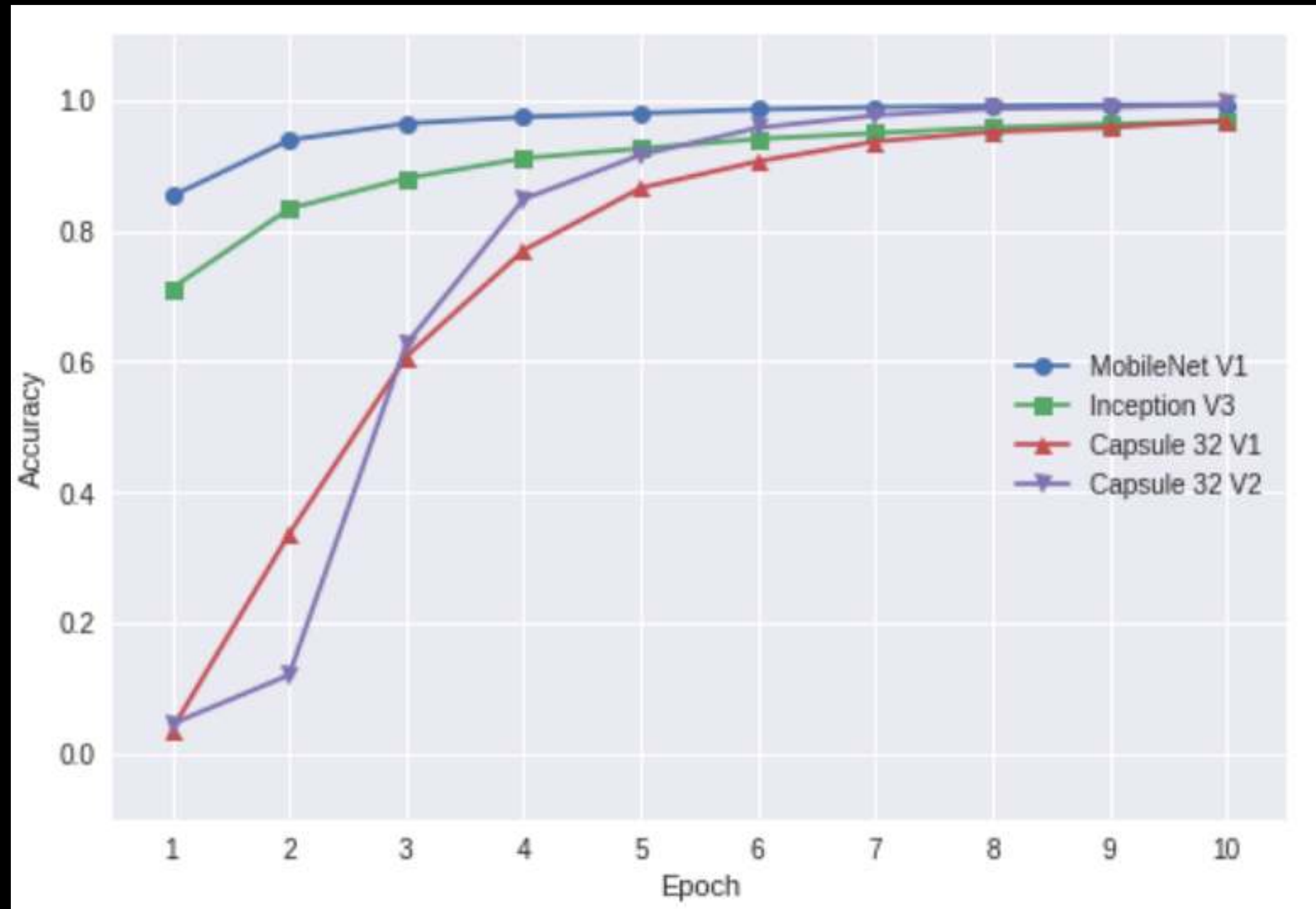


Automatic Sign Language Detection Task

* Sign language detection task



Comparing capsule networks with other architectures



Deep learning on graphs



Graph definition (Undirected)

✳ (V, E) denotes the vertices and edges in a graph

✓ $|V|$ - denotes the number of vertices

✳ $A = [a_{ij}]$ denote the adjacency matrix of the graph

✓ Similarity or affinity of the vertices.

✓ Symmetric and typically sparse matrix

✳ $D = \text{diag}[d_1, d_2, \dots, d_N]$ denote the degree matrix

$$d_i = \sum_j a_{ij}$$



Defining graphs

✳ Input features

$$\mathbf{x}_i \in \mathcal{R}^D \quad i = 1 \dots N$$

✳ Input feature space

$$\mathbf{X} \in \mathcal{R}^{N \times D}$$

✳ Hidden layer initialization

$$\mathbf{H}^0 = \mathbf{X}$$



3-steps in Graph convolutional networks

✳ I. Feature propagation

$$\bar{\mathbf{h}}_i^k = \frac{\mathbf{h}_i}{d_i + 1} + \sum_{j=1}^N \frac{a_{ij}}{\sqrt{(d_i + 1)(d_j + 1)}} \mathbf{h}_j^{k-1}$$

$$\mathbf{S} = \tilde{\mathbf{D}}^{-\frac{1}{2}} \tilde{\mathbf{A}} \tilde{\mathbf{D}}^{-\frac{1}{2}},$$

$$\bar{\mathbf{H}}^{(k)} \leftarrow \mathbf{S} \mathbf{H}^{(k-1)}.$$



Graph convolutions

✳ Non-linearity and activations

$$\mathbf{H}^{(k)} \leftarrow \text{ReLU} \left(\bar{\mathbf{H}}^{(k)} \mathbf{\Theta}^{(k)} \right)$$

$$\hat{\mathbf{Y}}_{\text{GCN}} = \text{softmax} \left(\mathbf{S} \mathbf{H}^{(K-1)} \mathbf{\Theta}^{(K)} \right)$$

